

Mars Reconnaissance Orbiter Aerobraking Navigation Operation

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After a seven-month interplanetary cruise, the Mars Reconnaissance Orbiter arrived at Mars and executed a 1.0 km/s Mars Orbit Insertion (MOI) maneuver. The post-MOI orbit was highly elliptical with a 35 hour, 428km x 45000km altitude orbit. To establish a useful science orbit, the navigation team used an aerobraking technique to guide the spacecraft into a 2-hour, 255km x 320 km altitude orbit. This paper details the aerobraking navigation operation strategy and flight results. It also describes the aerobraking key requirements and navigation challenges.

I. Introduction

The Mars Reconnaissance Orbiter (MRO) was launched onboard an Atlas V-401 from Cape Canaveral Air Station on August 12, 2005. After a seven month cruise, MRO arrived at Mars on March 10, 2006 and performed a Mars orbit insertion (MOI) maneuver. This maneuver placed MRO in a highly elliptical orbit with an initial orbit period of 35 hours, an apoapsis altitude of approximately 45,000km and a periapsis altitude of 430km. In order to save fuel, an aerobraking technique was used to lower the orbit and establish the desired science orbit. The focus of this paper is on the navigation operations strategy during aerobraking, the key requirements, the challenges the team encountered, and the results and lessons learned.

A. Aerobraking Overview

The aerobraking technique required the use of the atmospheric drag to reduce the spacecraft velocity and orbit period. Figure 1 below shows the aerobraking effect on the MRO orbit. During the six month aerobraking phase, the apoapsis altitude reduced from 45000km to 486km saving about 1.2 km/s and the local mean solar time (LMST) drifted from 8:30PM over to 3:00PM as desired. Aerobraking maneuvers (ABM) were required at apoapsis to either lower or raise the periapsis altitude in order to either maintain sufficient drag on the spacecraft (measured in heating rate), to avoid possible collisions with the other three Mars orbiters (Mars Global Surveyor, Odyssey, Mars Express), or to maintain a 2 day lifetime per the project requirement. An overview of the aerobraking timeline is shown in Figure 2. Prior to starting aerobraking, MRO stayed in a near drag-free altitude (periapsis altitude at 426km) for 20 days for system and sequence checkouts.

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There were four sub-phases to the aerobraking phase. The Walk-in sub-phase started on March 30, 2006 and lasted about two weeks. During this sub phase, six aerobraking maneuvers (ABM) were performed to slowly lower the periapsis altitude from 430km to 108 km into the Martian atmosphere. All ABMs were selected from the navigation developed ΔV menu. There were 20 ΔV choices in the menu ranging from 0.03 m/s to 4.2 m/s¹. The Main sub-phase lasted about four months and consisted of 13 maneuvers used mainly for corridor control to maintain sufficient drag. The majority of the apoapsis reduction occurred during this time. The Walk-out sub-phase lasted about one week and consisted of 7 maneuvers for both collision avoidance and maintaining the 2 day lifetime constraint. When the apoapsis altitude reached about 485 km, the aerobraking termination maneuver (ABX) was performed on August 30, 2006 at which point the aerobraking phase ended and the transition phase began. Five maneuvers were required over the next four months during the transition phase in order to establish the desired science orbit. Table 1 summarizes the ABM status. A total of 44.1 m/s were used in aerobraking including 14.5 m/s for walk-in maneuvers, 3.4 m/s for corridor control, 1.2 m/s for COLA, and 25 m/s for termination. There were 19 down maneuvers and 8 up maneuvers.

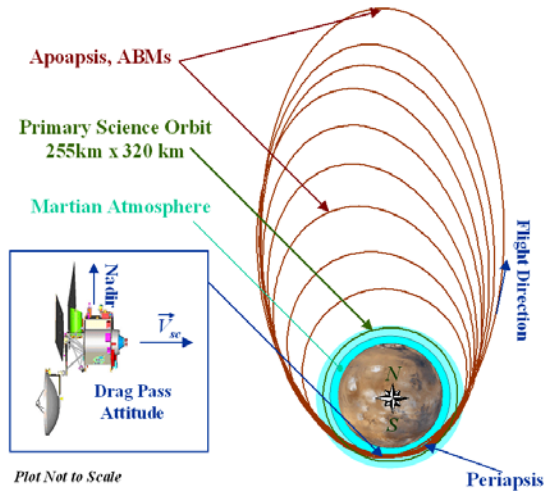


Figure 1. Aerobraking Operation Concept

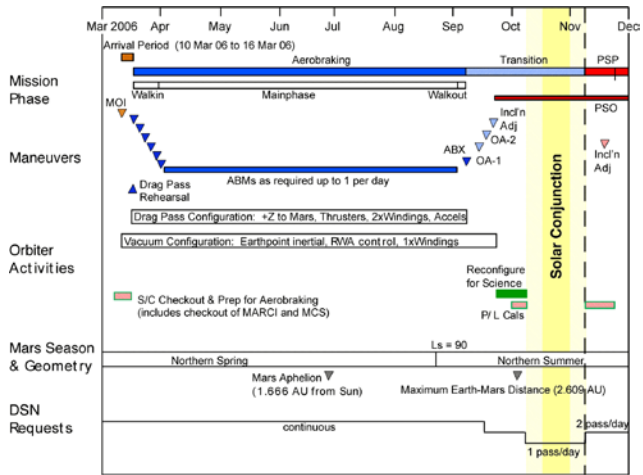


Figure 2. MRO Aerobraking Timeline

Table 1: Aerobraking Maneuver Summary

ABM Type	Number of ABM	Total DV, m/s	Composition
Walk-in	6	14.5	6 Down
Corridor Control	14	3.4	4 Up, 10 Down
COLA	6	1.2	3 Up, 3 Down
Sub-Total	26	19.1	7 Up, 19 Down
ABX	1	25	1 Up
Total	27	44.1	8 Up, 19 Down

B. Spacecraft Operations Overview

Three spacecraft attitudes were used during the aerobraking phase. An earth-pointed attitude was used while the spacecraft was in the vacuum portion of orbit. In this configurations as shown in Figure 3, the spacecraft high gain antenna (HGA) and solar panels were in a spread eagle configuration with the gimbals in power hold. The spacecraft -Y axis was pointed to Earth, and the +Z axis was in the directions of the Earth vector crossed with the Sun vector. The second attitude was the drag pass configuration. The HGA and solar panels were in the spread

eagle position with primary and secondary gimbals in power hold. The spacecraft +Y axis was aligned with the velocity vector and the +Z axis was pointed down towards Mars as shown in Figure 4. During the drag passes, there was active thruster control to maintain “loose” deadbands. The total drag area in this configuration was 37.5 m². The third attitude was the ABM configuration for use during maneuvers at apoapsis. This configuration is the same as the vacuum configuration except the +Y axis is either along the velocity vector or opposite the velocity vector (depending on if the maneuver lowers the periapsis altitude or raises it) and the +Z axis is pointed down towards Mars.

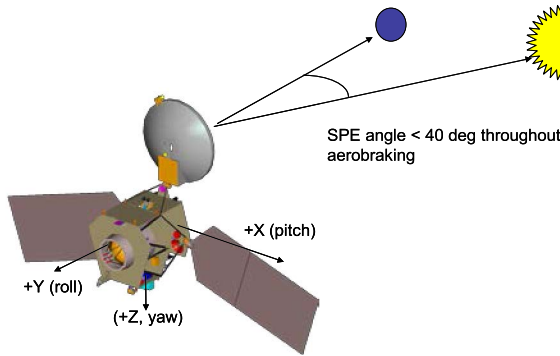


Figure 3. MRO Aerobraking Vacuum Configuration

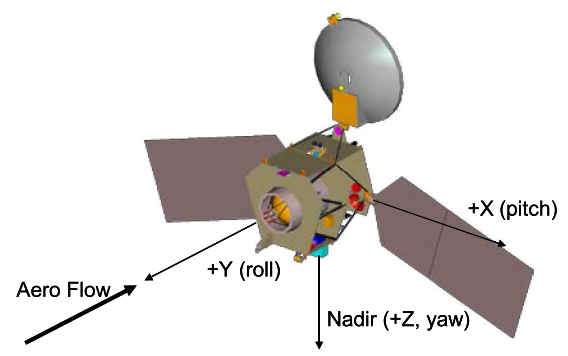


Figure 4. MRO Drag Pass Configuration

II. Navigation Aerobraking Interface and process

A. MRO Aerobraking process description

The navigation process consisted of two major sub-processes: Orbit Determination (OD) and Orbit Control & Trajectory Analysis. Each sub-system component contained dozens of programs.

1. Orbit Determination Process

Orbit Determination was one of the core elements in the Navigation Process. As shown in Figure 5, it started with a full range of information (data and models) collection. Real-time engineering data including reconstructed spacecraft attitude and on-board small forces due to momentum desaturation were generated through the telemetry query system. Along with the tracking data and other ancillary information provided by the Ground Data System, the orbit determination process was performed through detailed modeling and the spacecraft dynamic model fine-tuning, data arc and ancillary data setup, and estimation strategy update.

The nominal spacecraft ephemeris and partials were generated via trajectory analysis processor - Jet Propulsion Laboratory's (JPL's) Double Precision Trajectory (DPTRAJ) and followed by observable and data reduction process

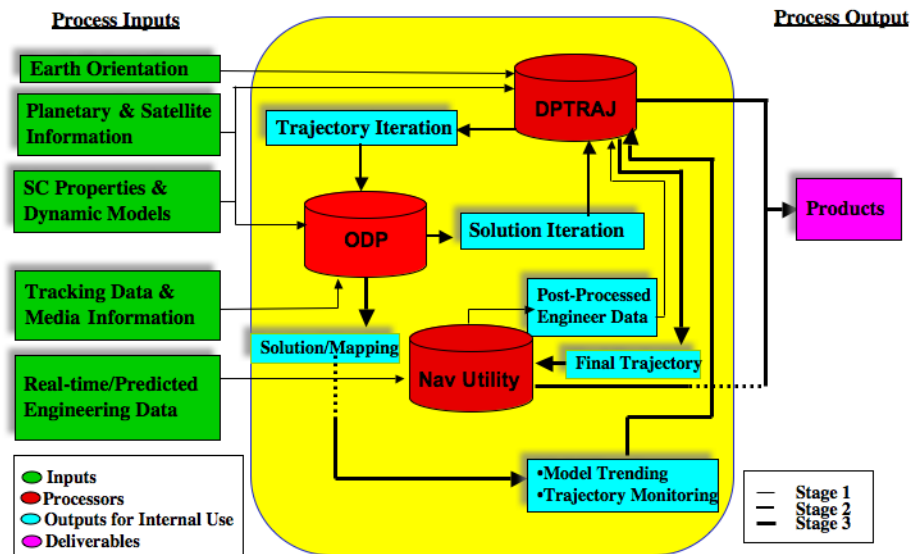


Figure 5: Simplified Navigation Process

- Orbit Determination Program (ODP). The orbit solution was generated for trajectory and model iterations. The byproducts such as reconstructed maneuver/small forces/solar radiation parameters, tracking data residual, and solution statistic were generated to perform dynamic calibrations, tracking data analysis, and covariance analysis. The converged OD solution could also be propagated to any future time for trajectory product generations. The outputs then were used by other systems such as Science Planning and Sequence Development. The Navigation team performed orbit determination daily during nominal operation and provided continuous support for the critical events.

2. Orbit Control Process

The Orbit Control design played an important role in the process. It was responsible for the reference trajectory generation and maneuver design. The resulting design was incorporated in the spacecraft dynamic model. Integrated spacecraft ephemeris with maneuver(s) were used for trajectory analysis, planning, and product generation.

Orbit Control Maneuvers traditionally were designed to trim the orbit for specific mission objectives. For instance, an Aerobraking Maneuver (ABM) was for aerobraking corridor control, a walk-in maneuver was to initiate aerobraking orbits, and a walk-out maneuver was to terminate aerobraking orbits. The typical 10-day maneuver design process was not suitable for all these maneuvers. A pre-canned maneuver (both burn attitude and magnitude) strategy was adapted for the ABMs to generate a quick turn-around product. An ABM could be selected hours before execution.

3. Sequence of Events Process and PTE

The sequence team would build the drag pass sequences and the ABM sequences. The drag pass sequence contained absolute time-tagged events and was built with the Automated Sequence Processor (ASP) utilizing the Nav predicted OPTG file (containing periapsis times and max dynamic pressure delta time from periapsis) and the Autogen Parameter File (APF) (containing weekly reset drag pass block parameters). Figure 6 below details the drag pass sequence of events.

The sequence team also built the ABM sequence with the Automated Sequence Processor (ASP) utilizing the Nav predicted OPTG file (containing the apoapsis time for the desired orbit), the Autogen Parameter File (APF) (containing the Weekly Reset ABM Block Parameters, the desired burn direction (“up” to raise periapsis or “down” to lower periapsis), the desired ΔV magnitude chosen from pre-validated menu of ΔV options. The mars ephemeris file was used to derive the desired S/C burn attitude to accomplish the “up” or “down” maneuver. Figure 7 below details the ABM sequence of events.

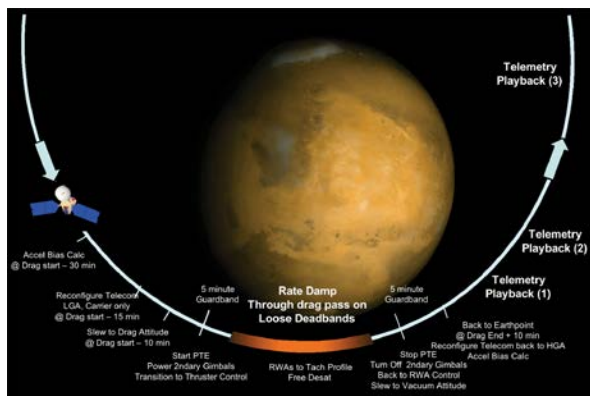


Figure 6. Drag Pass Sequence of Events

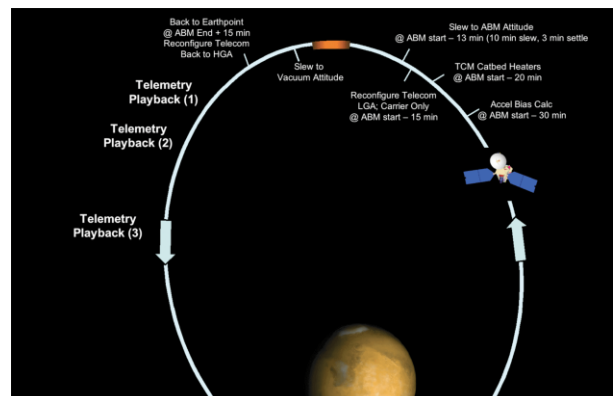


Figure 7. ABM Sequence of Events

One of the main requirements for the navigation team was for the predicted periapsis times to be within 225 seconds of the actual times. As described above, the sequence of events were built based on the predicted periapsis time from the navigation OPTG file to determine when the spacecraft needed to slew into a safe drag pass configuration and when it was safe to slew back to the vacuum pass configuration for communication with Earth. If the predicted periapsis time was off by more than 225 seconds from the actual periapsis time, the onboard sequence would guide the spacecraft into the drag pass configuration either too soon or too late, possibly leading to spacecraft damage. However, predicting the periapsis times accurately was a difficult task due to the uncertainty of the atmosphere density.

The MRO mission used the Periapsis Timing Estimator (PTE) developed by LMA and used previously on the Odyssey mission. PTE performed on-board calculations of the periapsis timing error using the accelerometer data. The sequences were built to incorporate PTE results and adjusted the ephemeris accordingly to keep spacecraft attitude consistent with the updated timing. With the use of PTE, the number of orbits per sequence was no longer constrained by navigation's ability to predict periapsis timing to within 225 seconds. The PTE algorithm was updated for MRO to improve the accuracy and robustness. The algorithm used the accelerometer data to compute actual drag pass DV in addition to the time of peak acceleration. The actual DV was used to compute the actual orbit period change. Both the period change and timing error were used to adjust subsequent sequence timing.

B. Navigation Roles and Responsibilities

The main navigation tasks included orbit determination, trajectory analysis, and maneuver design. The navigation team worked closely with the other project teams to insure a smooth flow of information. The navigation team was responsible for producing the following output files for the other project teams; light time files (LTF), Orbit Propagation and Timing Geometry File (OPTG), trajectory file (SPK or P-file), maneuver profile file (MPF),

The team would perform the following tasks on a weekly basis; trend and analyze the atmosphere model, run out the trajectory and compare to the baseline, make adjustments to the heating rate corridor as needed, select the appropriate DV maneuvers for the week as well as the pop up maneuver and immediate action maneuver, collision avoidance analysis, and an independent team at the Langley Research Center (LaRC) would perform thermal and trajectory analysis. The navigation team would present an update each week at the project weekly reset meeting held every Wednesday at 9 am (PDT).

The team would perform the following tasks on a daily basis; orbit determinations, trend and analyze the atmosphere model, run out the trajectory and compare to the baseline, analyze possible maneuvers if required, collision avoidance analysis, and the LaRC team would also analyze the trajectory and possible maneuvers. The navigation team would present an update and maneuver recommendation everyday at the Aerobraking Planning Group (APG) meeting at 1 pm (PDT).

There was also an Atmospheric Advisory Group (AAG) that met daily at 11 am (PDT) to discuss the latest atmosphere trends. A navigation team member would attend this meeting to provide a status and gather information on the latest atmosphere trends. The AAG gathered information from the MGS and ODY instruments for atmospheric monitoring. They also used the high rate accelerometer data recorded onboard MRO throughout each drag pass for atmospheric science. From this, they generated trends in the atmosphere and reported back to the project at the APG meeting.

III. Navigation Aerobraking Strategy

A. Design Principles

The aerobraking baseline flight profile was designed to meet all of the project systems requirements and achieve the desired science orbit. Prior to launch, the following navigation related requirements were developed for aerobraking operations². These requirements were built into the overall aerobraking design and process.

- The Project System shall support aerobraking for lowering the capture orbit apoapsis to an altitude of 450 km
- The PS shall be designed consistent with a LMST of 3:00 pm +15/-15 minutes.
- The Project System shall be capable of supporting 180 days of continuous aerobraking
- The Project System shall accommodate an orbit-to-orbit atmospheric variability of 90%
- The Project System shall have the capability during aerobraking to perform, on any orbit, daily maneuvers designed to adjust the periapsis altitude
- The Project System shall be designed to survive for 48 hours in aerobraking
- The Project System shall predict aerobraking periapsis uncertainty to less than 225 seconds for periapsis passage time (3s)
- The Project System shall be capable of performing aerobraking trim maneuvers (ABM) as frequently as once a day during the aerobraking phase
- The Project System shall be capable of performing at least one contingency pop-up maneuver to raise periapsis out of the atmosphere

Two of the major design constraints during aerobraking was safely reducing the apoapsis altitude from 45000km to 450km and achieving a 3pm LMST prior to the start of solar conjunction on October 7, 2006. Other contributing

design constraints including building in margin at the front end of aerobraking to be available towards the end, power and thermal constraints, collision avoidance, inclination control, and maintaining the apoapsis decay to 300km or less over 48 hours with no more than one maneuver per day during the end game.

At the start of aerobraking the LMST was at 8:30pm which is a result of the launch/arrival period. The LMST drift from 8:30 pm to 3:00 pm was a result of the motion of Mars around the Sun. It takes about six months for Mars to rotate around the Sun at about 1 deg/day (or 1hr of LMST drift per month) to achieve the 3:00 pm LMST target. At the end of aerobraking, the LMST flattens out when the orbit becomes Sun-Synchronous. The aerobraking baseline had to be designed such that the apoapsis altitude would decrease to 450 km in the same amount of time that it would take the LMST to drift over to 3:00 pm. The design of the apoapsis altitude decay had to also meet the requirement of being able to accommodate an orbit-to-orbit atmosphere variation of 90% by providing at least 150% margin at all times.

During the walk-out phase, the main concern was the requirement that the spacecraft must be able to survive 48 hours of aerobraking. During the walkout phase, the gravitational perturbations will naturally drive periapsis out of the atmosphere. In order to stay on course and finish in the proper orbit, the baseline trajectory design had planned maneuvers such that periapsis would be driven back into the atmosphere but not so far that the 48 hour lifetime requirement could be violated and without violating the ERD Heating Rate Limit.

1. Mid Course Design Update

The original baseline trajectory required aerobraking until 9/11/06. In mid-June, halfway through aerobraking, the project decided to move the aerobraking end date up by two weeks in order to allow for additional time for instrument checkout before solar conjunction, to take some critical HiRISE observations of the proposed Phoenix landing site, and to avoid DSN scheduling conflicts with the STEREO mission launch. The navigation team was able to redesign the overall aerobraking profile but the new LMST target had to change to 3:10 pm. This changed the overall strategy slightly. The original plan was to target a 3:15pm LMST (higher heating rates) for the first half of aerobraking and then back off and target a 3:00pm LMST for the second half of aerobraking. With the new end date, the second half of the aerobraking phase had to maintain slightly higher heating rates in order to finish early but still maintained healthy thermal margins for spacecraft safety.

B. ABM decisions and recommendations

There were four types of ABMs available for use during aerobraking. The only type of maneuver actually used during MRO aerobraking was the Corridor Control ABM. Nominal Corridor Control ABMs were used to keep the spacecraft within the desired periapsis altitude corridor and maintain progress with respect to the glideslope. They were non-time critical maneuvers scheduled for execution at apoapsis. Typical execution frequency was one every 2 weeks during the main phase and one per day during the end game. These maneuvers were selected from the menu of 20 pre tested maneuvers ranging from 0.03 m/s to 4.2 m/s. The spacecraft calculated the proper maneuver attitude on-board, the +Z axis would be nadir-pointed and the +Y axis would be either along or opposite the velocity vector.

The other three maneuver types included:

An Immediate Action ABM was to be executed in response to a solar array temperature derived Qdot limit violation seen immediately after a drag pass from telemetry, or in response to the Atmospheric Advisory Group (AAG) determination of a dust storm from the MGS TES or Odyssey THEMIS data. The maneuver was always to be in the “up” directions and would raise the periapsis altitude by at least 7 km. The onboard PTE software had the capability to initiate this type of maneuver but was never enabled. This type of maneuver was not required during MRO aerobraking.

A Manual Pop-up maneuver was to be initiated by the ground in response to an unsafe spacecraft anomaly. The ΔV Magnitude and burn attitude were determined weekly to ensure periapsis would be raised to 150 km. At this altitude, the spacecraft would be out of the detectable atmosphere and can remain on RWA control for the entire orbit.

The Autonomous Safe Mode Pop-up was to be executed autonomously by safe mode after attitude initialization (as required) and after defined number of orbits (nominally set to zero to execute at the first apoapsis opportunity). The ΔV Magnitude and burn attitude were the same as the Manual Pop-up maneuver. The fault protection Navigation Performance Monitor (NPM) was to derive the apoapsis time for the burn execution but was never fully tested and implemented in operations. While there were two safemode events and spacecraft anomalies, the situations did not warrant either a manual or autonomous pop-up maneuver.

C. Glideslope analysis and monitoring

As mentioned above, the original plan was to exit aerobraking September 11, 2006 and fix Local Mean Solar Time (LMST) at 3:00 pm (shown in Figure 8 labeled as “Original Baseline”). By executing a more aggressive aerobraking profile starting at the 4.5-hour orbit period (i.e. raised the corridor limits by about 15%), MRO concluded aerobraking two weeks earlier on August 30, 2006 after 445 orbits. The post-ABX orbit was 215km x 485km with 1.9 hours orbital period and 3:11 pm LMST. To better follow the glide-slope parameters (e.g. period, LMST shown in Figure 8) and meet the thermal (expressed in $\dot{q}$, W/cm^2) constraint, MRO evaluated the corridor every week. There were times that the corridor was adjusted weekly to accommodate the flight profile changes and atmosphere fluctuations. As shown in Figure 9, the actual flight profile (in black) was deviated from the original baseline (in green) at orbit 200 or so.

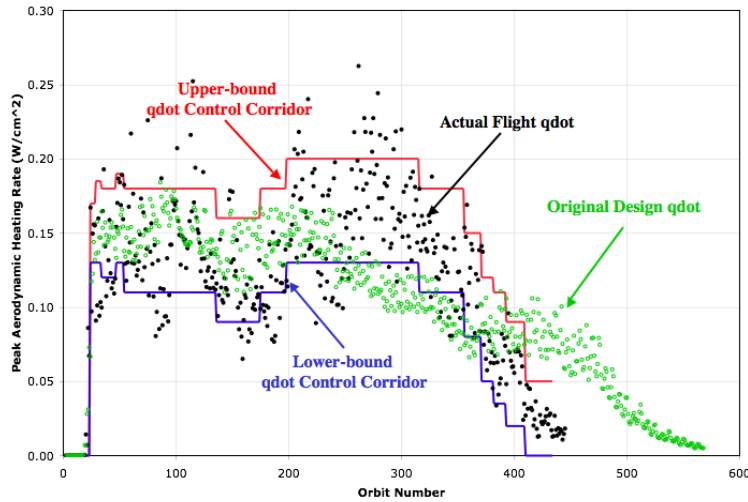


Figure 8: Aerobraking $\dot{q}$ and corridor

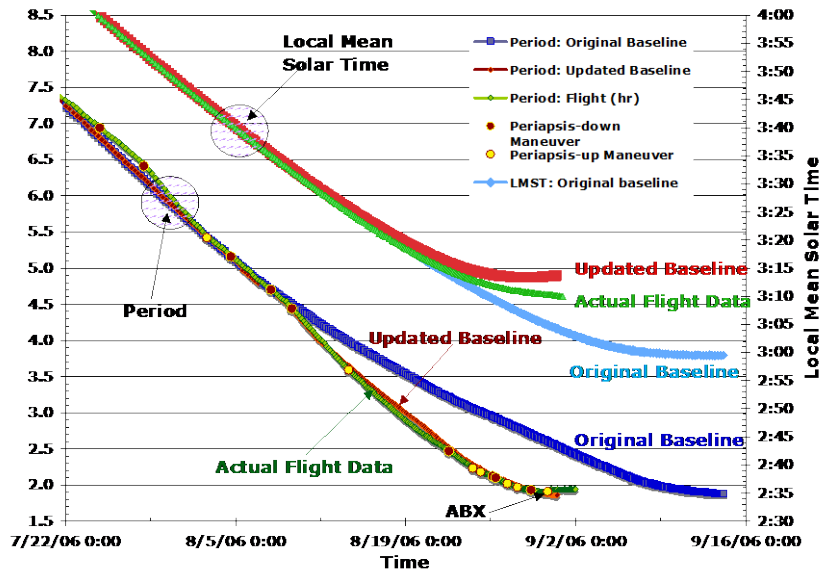


Figure 9: Period and LMST Glide Slopes during End-Game

At the beginning of aerobraking, the apoapsis reduction rate was in a range of several hundred kilometers per orbit, and it was much more sensitive to the atmosphere fluctuation. As the aerobraking progressed, the reduction rate decreased quickly and was less sensitive to the density uncertainty. During the end-game, the rate could reduce to less than 10 km per orbit. For a 2-hour orbit, the daily reduction capability was on the level of 100km. Similarly, the period change per orbit had the same trend (tied to the semi-major reduction). Clearly, to build up the glide-slope margin, it was required to brake harder as early as possible. However, it was the later portion of the aerobraking that provided better control. Figure 10 illustrates the capabilities of period reduction and apoapsis altitude reduction (per orbit) in relation to the period and drag- ΔV (per orbit). In the first four months, the average drag- ΔV was about 2 m/s and the drag-duration was ranging from 5-8 minutes per orbit. Toward the end, the drag- ΔV peaked at ~ 6 m/s. The drag-duration increased steadily to 20-minute level as the orbital period reduced to 2 hours.

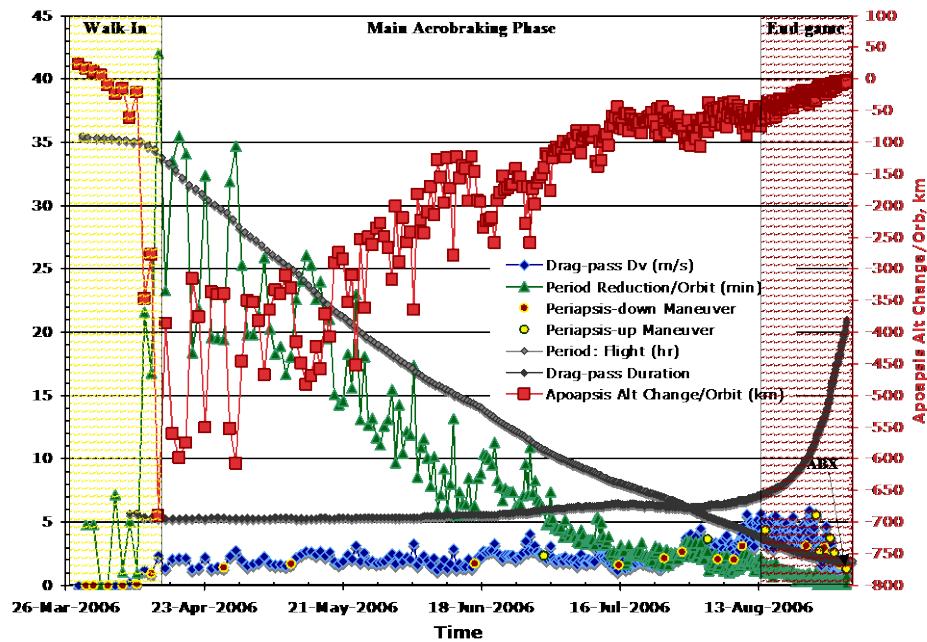


Figure 10: Drag-Pass ΔV & Duration, Period & Period Reduction, and Apoapsis Altitude Reduction

Using the aerobraking technique, the apoapsis altitude was reduced from 45,000 km to 485 km in 5 months, as shown in Figure 11. More than 50% of the ABM efforts during the end-game were dedicated to resolving COLA issues. Also, selection of the ABX orbit in the last week of aerobraking was especially difficult due to COLA issues.

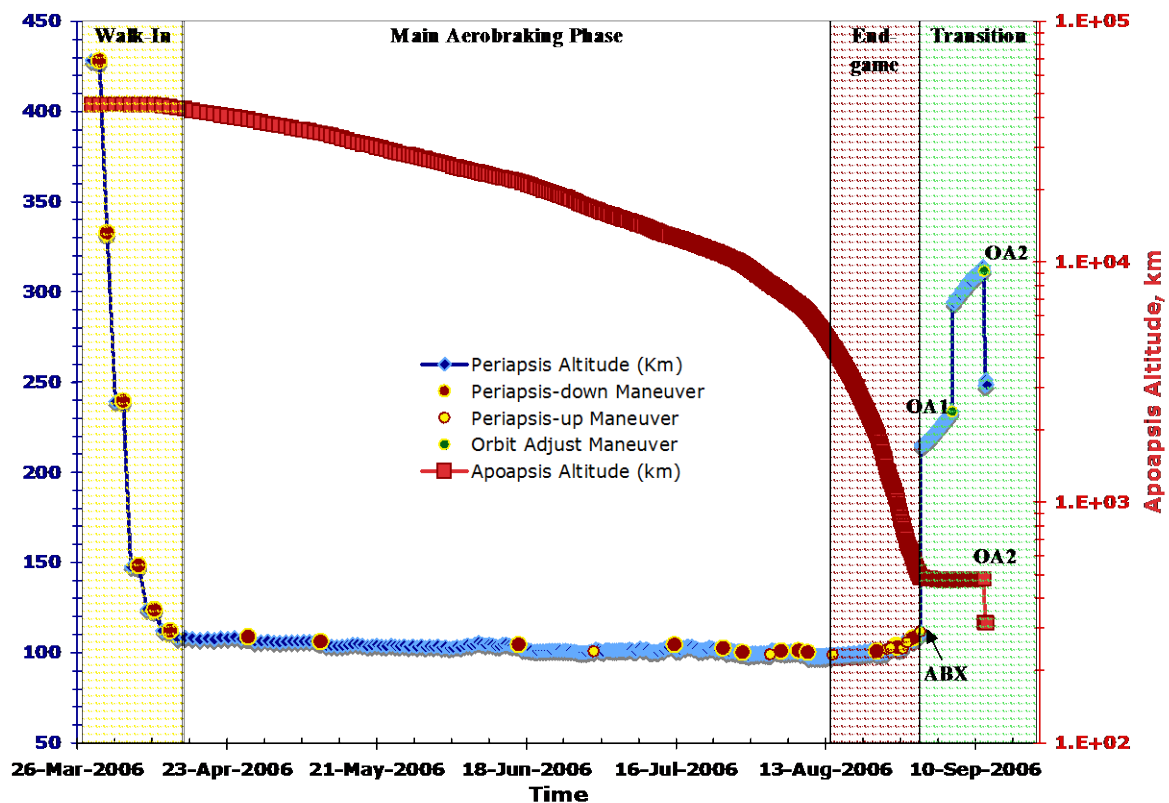


Figure 11: Aerobraking Altitude and ABM

The aerobraking termination maneuver (ABX) utilized the nominal ABM sequence block and implementation process. The ABX- ΔV of 25 m/s was pre-built and tested and was on board at the time of ABX execution. Twenty-four hours before ABX, the primary ABX orbit was selected along with an alternative orbit for contingency. The timing error was adjusted by the on-board “Periapsis Timing Estimator”, which compensated the delta period error induced by the atmosphere density mis-modeling. The ABX burn was an “up” maneuver (i.e. attitude configuration file), which was used in the ABM process throughout the aerobraking operation.

The navigation team monitored the ABX orbit conditions starting 7 days before the planned termination date. Key parameters in the watch list included the 48-hour orbit lifetime, ABX primary and backup orbits, and COLA conditions. To satisfy the orbit lifetime constraint, the navigation team was required to maintain an apoapsis altitude decay of 300km or less for at least 48 hours. One periapsis-up ABM was required and raised the periapsis altitude 0.5 km on 28 August 2006 to avoid violating the lifetime requirement.

IV. Navigation Challenges

A. Operation planning and implementation

1. Continuous Support

Aerobraking phase was a 24-hour / 7 days per week effort. The activities during this critical phase were supported mainly by the Navigation Team (Nav) and the Flight Engineer Team (FET). The MRO team inherited aerobraking processes from Magellan, Mars Global Surveyor, and Mars Odysseys. The processes were divided into strategic and tactical categories. Strategic operations were mainly interested in mid-long range planning, while tactical operations were concentrated on the short-term implementations. The strategic and tactical processes were coupled and highly correlated. The strategic processes provided the implementation parameters for the tactical processes, at the same time, the tactical processes feed back real-time information into the strategic process models. Table 2 summarizes the key aerobraking processes and their associated categorizations. The implementation of these

activities primarily required ground processes, such as drag sequence generations and validations, ABM design and commanding, verification of the uplink products, real-time monitoring, and atmosphere reconstructions. To support these activities, continuous DSN and uplink coverage and navigation operation was required to fulfill the tactical processes. Even though aerobraking was a gradual process, the risk increased as the orbit size decreased and the ability to respond to a contingency event decreased. This was an important factor for why the 24-7 monitoring mechanism needed to be in place in order to deal with the unexpected, such as COLA event.

Table 2: Aerobraking Strategic and Tactical Processes

Name	Objectives	Category
Atmospheric Modeling Process	To update the navigation models and coefficients	Tactical
Weekly Reset Process	To monitor progress against baseline trajectory; to analyze subsystem performance for the upcoming week's trajectory; to select parameters and nominal build orbits for the week; to OTB test drag & ABM sequences	Strategic
Daily Spacecraft Health Monitor Process	To monitor the health of the spacecraft; to trend the performance of the spacecraft	Tactical
Daily ABM Decision & Implementation Process	To assess whether an ABM is necessary to remain in the corridor; to build, review & uplink ABM sequences	Tactical
Mars Atmospheric Advisory Process	To trend the Mars atmosphere; to predict upcoming atmospheric trend including dust storm detection	Strategic/ Tactical
Nav & Sequence Update Process	To predict periapsis timing; to generate, review, and uplink drag sequences and spacecraft ephemerides	Tactical
Orbit-by-Orbit Spacecraft Health Monitor Process	To estimate DP and timing errors; to check for alarms/anomalies; to reconstruct thruster pulses; to trend critical spacecraft performance	Tactical
Immediate Action Process	To assess the need for emergency ABMs or drag sequence builds; to build, review, and uplink these emergency sequences if needed	Tactical

2. Operation Constraints

Per the mission requirements, a corridor-control ABM could be performed once per day on any orbit. However, the MRO team put a self-constraint of limiting to one ABM opportunity per week during normal working hours for most of aerobraking. This strategy was to reduce the workload of the flight operation teams and to maintain a stable operation environment. However, it imposed a great challenge to the navigation team in terms of trailing and maintaining the designed glideslope parameters. The residual errors due to the constrained ABM opportunity had to be accounted for in the next design cycle and incorporated in the corridor adjustments. Also, improving the short-term atmosphere predictability was one of the major focuses to offset the impacts from the constraints. The weekly ABM not only was used to take out the atmosphere errors experienced in the past orbits but also to account for the future trend. The constraint was relaxed to a daily opportunity during the end-game. On average, MRO performed one corridor control ABM every ten days, which was significantly less than the previous aerobraking missions.

With the aid from the on-board Periapsis Timing Estimator (PTE), a similar constraint was applied to the on-board ephemeris update opportunities. The update frequency was limited to 2-3 times per week. Although, this had released the burden from the flight operation teams, it did not reduce the workload of the navigation team. The navigation team was still required to support 24-hour / 7-day operation (24-7). Since aerobraking was a half-year intensive operation, the challenge was to maintain an adequate working environment and independent operability of the staff. Pre-aerobraking trainings, using simulated and actual flight data from the previous missions was given to all personnel to ensure the familiarization of the aerobraking process and procedures. Cross-function training within the team was also conducted. This ensured the non-interruption of the operation support due to personnel absences. To maintain high working spirit, a 5-day/40-hour working schedule was enforced as much as possible. The gaps

were filled with part-time or cross-trained personnel. Due to the ephemeris timing uncertainty, a 1-hour overlap during shift transition was scheduled during smaller orbits. Automated tools to book-keep the flight database and email/log system were adapted to reduce the workload and human errors. Figure 12 illustrates a 24-7 navigation support scenario. Orbit determination was performed continuously to support the atmospheric density scale factor modeling, trajectory and ABM analysis, and COLA monitoring. Trajectory analysis and ABM design were conducted during the prime time. Except the last couple weeks of the aerobraking, weekend support was mainly concentrating on monitoring the atmosphere and trajectory. Typically, there were no ABM opportunities scheduled during weekends.

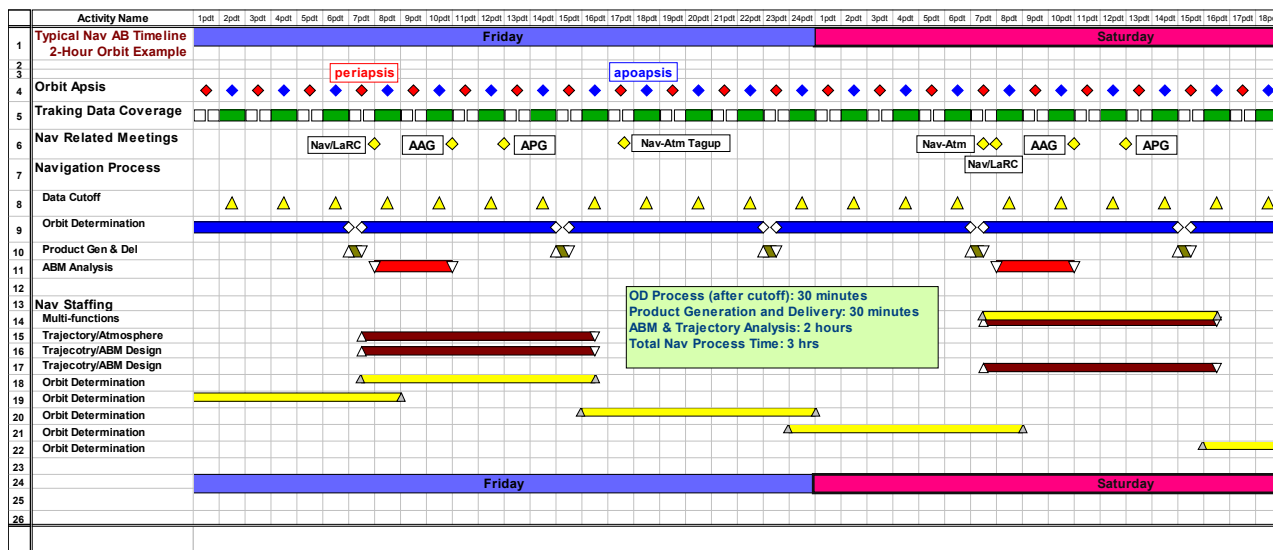


Figure 12: Navigation 24-7 Operational Scenario

B. Atmosphere modeling

1. Navigation Models and Filtering

Major navigation non-gravitational dynamic models include solar radiation pressure, which acts on the irregular-shaped spacecraft bus, and gimbal-enabled solar array and high gain antenna; acceleration resulting from the thermal imbalance; thruster firings occurring from the momentum buildup desaturation, attitude control, or any unexpected anomalies; any unanticipated outgassing; propulsive maneuvers implemented for trajectory/orbit control; and Martian atmosphere drag experienced at low orbit altitude.

Mis-modeling of the solar radiation pressure, thermal venting, and thruster events were the dominant error sources during interplanetary phase. In the aerobraking phase, MRO was at a low orbit altitude and atmosphere drag becomes a significant factor contributing to the navigation error.

Two-way X-band Doppler was the only data type used during aerobraking. The Doppler observables from the DSN tracking sites were modeled using the differenced range formulation in the DPODP. The transformation of the location of the tracking station from body fixed to inertial coordinates included polar motion calibrations and UT1-TAI timing corrections, solid Earth tides, and Earth center of mass correction. In addition to the seasonal corrections, the diurnal troposphere and ionosphere calibrations were also included in the computation. The Earth Orientation Parameter (EOP) data file was the source of the polar motion and timing corrections. To reconstruct the Martian atmospheric density every orbit and to possess the capability of performing uplink at any time, continuous Deep Space Network (DSN) tracking was maintained during aerobraking phases.

Once the observed tracking data were differenced with the predicted values based on the dynamic and observational models described above, filtering was done to minimize the resultant residuals in a least-squares sense by adjusting some subset of the model parameters. The aerobraking filtering strategy was a fairly standard one used on most missions. The only estimated non-gravitational parameter was atmospheric drag.

2. Atmospheric Density Model and Performance

The navigation team reconstructed every drag pass for the purpose of trending the atmospheric density scale factor. A predicted density scale factor, which sometimes included wave parameters, was applied to the navigation

baseline atmosphere model, MarsGRAM2005, for trajectory propagation. Reference 3 provides a detailed description of the MRO density scale factor. As shown in Figure 13, the aerobraking started at the southern hemisphere at -72 degrees periapsis latitude and migrated toward south. It reached the South Pole 3.5 months after walk-in. In this time span, the atmosphere density scale factors were trending up from an average of 3 to 4.5. After passing the South Pole, the periapsis precessed northward toward the equator and reached it in 1.5 months. The density scale factor decreased to an average of 1.2 at the equator. The MarsGRAM2005 model had the largest errors at the South Pole and the smallest errors near the equator. This phenomenon was analyzed a few weeks after the spacecraft turn-around the South Pole. A decaying rate was applied to a constant scale factor to reflect the observations. The challenge was to keep up the rate since the rate change is not a linear relationship and it had a second-order effect. Throughout aerobraking, the density scale factor running-mean for 5, 10 and 25-orbit were monitored. These reference trends showed how good the predictions were.

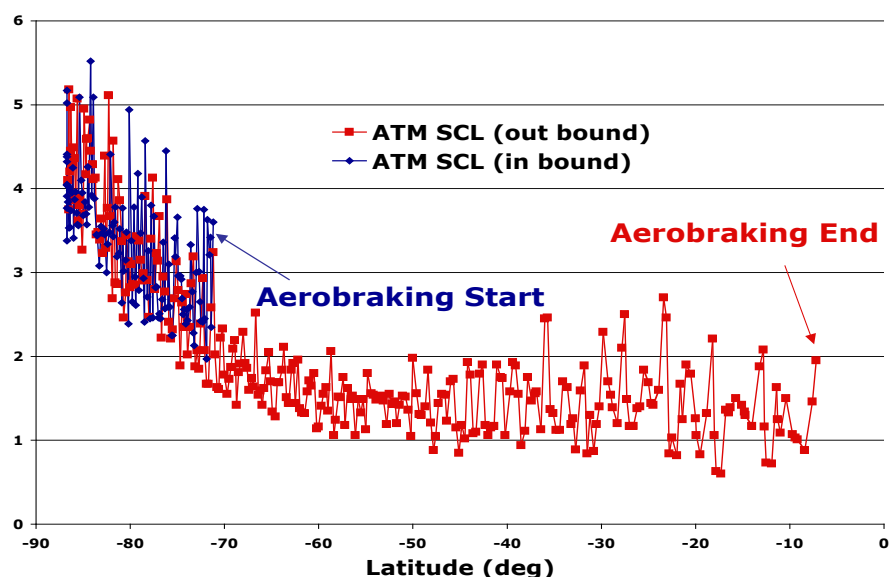


Figure 13: Reconstructed Density Scale Factor w.r.t Periapsis Latitude

Near the end of the aerobraking, the simple constant and rate formulation was no longer sufficient enough to model the rapid change of the atmosphere environment. A longitude-dependent wave function was introduced to better model the density scale factor. The switch was made based on strong evidences of the correlation in between the density scale factors and longitude. Post-aerobraking analysis indicated that the longitude-dependent wave model played a significant role in reducing the apoapsis altitude uncertainty. Terminating aerobraking at the desired altitude was a very important issue in dealing with post-aerobraking COLA issue.

Many utilities tools were developed for trending the reconstructed density scale factors. The tools output the statistical information, constant bias, rate, and longitude dependant wave coefficients. The atmosphere model was updated several times a week.

Figure 14 illustrates the density scale factor prediction performance. The error was computed by differencing the actual and predicted density scale factor and dividing by the predicted scale factor. The performance was expressed in terms of percentage. Positive values were under-predictions and negative were over-predictions. During aerobraking, the largest prediction error was about 70% under-predict. One-sigma performance was at 17% with mean near 0%. This indicates that the navigation team was able to predict perfectly the near-term atmosphere bias. The orbit-to-orbit errors were much less than the anticipated 35% uncertainty, 1-sigma. For the short-term prediction errors, the 5-orbit statistic was at about 30% level at the beginning of aerobraking and towards the end but less than 20% for most of the aerobraking. Similar signatures were presented in the 10-orbit and 25-orbit standard deviations except they were smoother with less local variations.

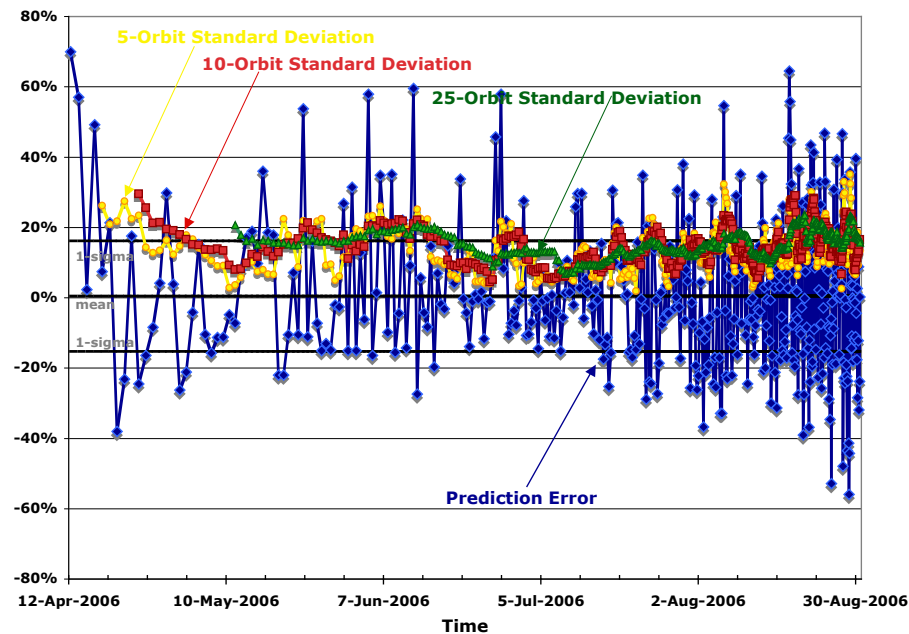


Figure 14: MRO Atmospheric Density Scale Factor Prediction Performance

3. Atmosphere update process

Atmosphere modeling was the major error source of the navigation dynamic models in aerobraking. It directly impacted the navigation timing accuracy for the on-board ephemeris. It also influenced the trending of the glideslope parameters and ABM determination. Without good understanding of its behavior and the associated environmental changes, an erroneous update might cause an unnecessary ABM execution or a false trajectory event (e.g. COLA).

Before updating an atmospheric scale factor to the baseline model, the navigation team investigated thoroughly the following factors: 1) Periapsis Altitude: the altitude could vary a few hundred meters from orbit to orbit but it could also change a few kilometers in a few days due to gravity influences. The atmospheric environment and glideslope parameters are usually affected by a large change to this parameter. 2) Density Scale Factor Analysis: various density scale factor trends were investigated such as linear/quadratic/exponential curve fittings. The objective was to ensure a model that could best represent the historical data and environmental variations. 3) Longitude Effect: due to the resonant effects, the spacecraft could visit the same neighboring longitude in a short period of time. From the observations, the density scale factor may have shown a high repeatability at the similar longitude region. This was especially true at the latter part of the aerobraking operation when the data points were abundant. 4) Latitude Effect: this had more of an effect on the mid-term predictions. The density model is correlated to the latitude. Usually, the error is present in a more gradual trend than an immediate change to the model. 5) Weather Report: the navigation team participated in the Atmospheric Advisory Group daily tag-ups and obtained the daily Martian weather report. One of the important items was the local and global dust storm analysis and forecast. An early warning of such an event could have significantly reduce the risk of spacecraft safety. There were a few local storms observed during aerobraking but none of these had threatened the spacecraft health or impacted the nominal atmosphere model.

Daily meetings were conducted to determine if it was necessary to update the model for future usage. The navigation team typically did not react to a single point of atmospheric data. Consistency was the main guideline on updating the atmosphere model. One of the goals was to maintain the bias within 10% of the running mean. On average, the navigation team updated atmospheric scale factor every 6 days. The predicted performance was about 17%, 1-sigma. This significantly helped to improve the aerobraking operation. Compared with the other aerobraking missions, MRO was much more efficiency in terms of trajectory control.

C. Collision Avoidance (COLA)

During the aerobraking process, it was possible that MRO would cross the orbital paths of the three other operational spacecraft orbiting Mars (Mars Global Surveyor (MGS), Odyssey (ODY), Mars Express (MEX)) or the two Martian moons (Phobos and Deimos). Prior to the start of aerobraking, the navigation team performed a preliminary COLA analysis, focusing only on the orbit crossing distances as shown in Figure 15 below. The MRO aerobraking baseline trajectory was used for the analysis along with the long term prediction trajectories for MGS, ODY, and MEX, and long term predictions for Phobos and Deimos. From this analysis, time frames were identified for when the MRO baseline trajectory was predicted to cross the orbital paths of MGS, ODY, MEX and Phobos. The COLA “seasons” seem to be correlated to the relative orbit geometries. As long as the navigation team kept the trajectory close to the baseline, we knew when to be aware of potential COLA issues.

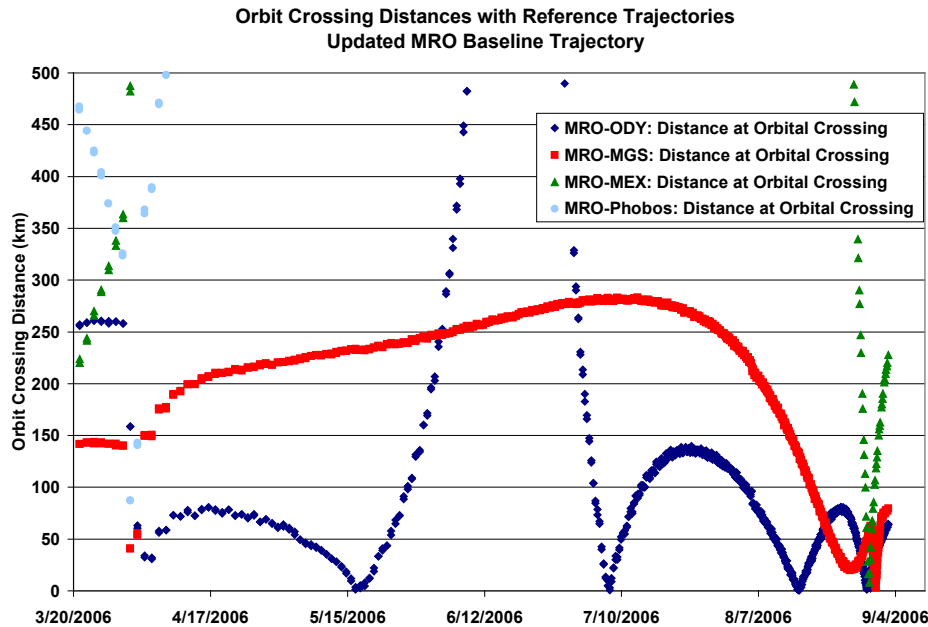


Figure 15. Aerobraking COLA Analysis with update MRO Trajectory

The navigation team developed a plan to minimize the possibility of a collision during these time frames. In general, the plan consisted of using the predicted trajectory files and looking at the radial distance between the orbits. If the radial distance between the orbits was determined to be within the MRO 5 sigma radial uncertainty, then the downtrack distance between the bodies was analyzed. If the downtrack distance was determined to be within the MRO 5 sigma downtrack uncertainty, a collision avoidance maneuver was then considered. A flowchart of the process is shown below in Figure 16 and more details about the strategy and how it was developed and implemented can be found in Reference 4. This was the first mission to implement such a process for use at Mars.

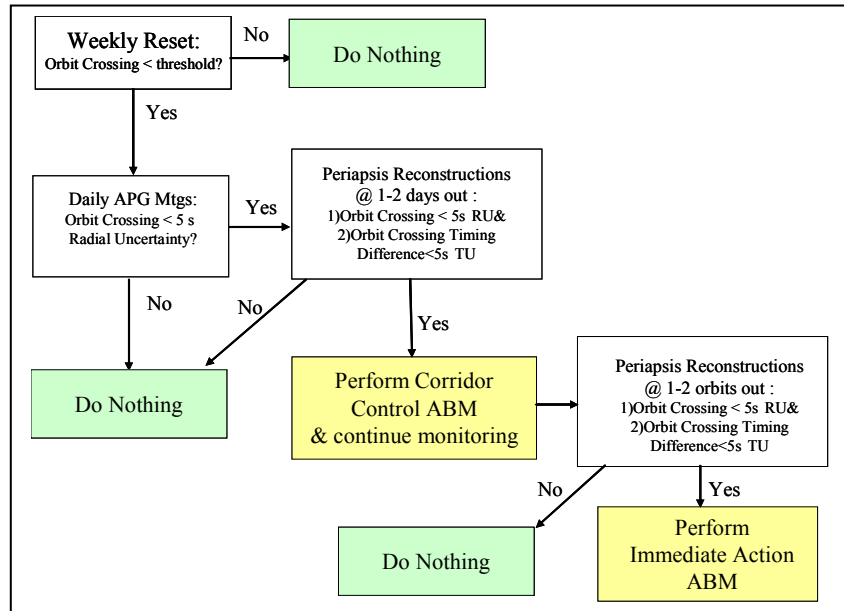


Figure 16. MRO Collision Avoidance Process

E. End Game

As shown in Figure 15 above, the majority of the COLA “seasons” were during the walk-out sub phase. There were also COLA issues right after the planned ABX maneuver which had to be taken into account while designing the maneuver. All of these COLA issues made the last two weeks of aerobraking very interesting for the navigation team requiring the team to work 24-7. During this time, there were about 12 orbits per day and an orbit determination solution was made for each one. The COLA situation was also analyzed for each orbit and the ABX maneuver analysis was updated on a daily basis.

A total of six maneuvers were implemented for COLA purposes during the last two weeks of aerobraking. These were small corridor control maneuvers executed a few orbits before the COLA event in order to tweak the orbit slightly out of either the 5 sigma radial or 5 sigma downtrack uncertainty range.

Selecting an orbit for ABX was quite a complicated process. The ABX orbit needed to satisfy the following criteria: (1) LMST at ABX of 3:10PM $\pm$ 5 minutes; (2) ABX altitude greater than 450km and less than 600km; (3) Eliminate potential post-ABX orbit collisions with other Mars orbiters. Two ABX opportunities, orbit-445 and orbit-447 were selected on August 29, 2006. At that time, resolving the post-ABX COLA issues was the driver in selecting these ABX orbits. To successfully avoid the post-ABX COLA issues, the ABX apoapsis altitude needed to be in the range of 465- 490km (see Reference 4). The predicted apoapsis altitudes were 480 km for orbit-445 and 470 km for orbit-447. For a 24-hour prediction period during the walk-out phase, the apoapsis altitude prediction error could be more than 50km due to atmosphere uncertainty. An atmosphere prediction model with complex longitude wave coefficients was used in the final delivery (see Reference 3). The result was exceptionally good. Based on the reconstructed solution, the apoapsis altitude of ABX orbit-445 was at 486km. It was only 6 km off from the prediction generated a day before. This marked a perfect ending to the five-month aerobraking operation.

V. Summary/Conclusions

MRO concluded aerobraking on August 30, 2006 after 445 orbits. The post-ABX orbit was 215km x 485km with 1.9 hours orbital period and 3:11 pm LMST. Five maneuvers were required over the next four months to tweak the orbit into the desired primary science orbit⁵. Overall, aerobraking saved 1.2 km/s ΔV .

Mars Reconnaissance Orbiter is the first mission in a low Mars orbit to carry high precision imaging instruments along with other science and experimental engineering payloads. The significant amount of science data return has helped us gain more knowledge and understanding of our neighboring planet. This paper presented the navigation operation activities and results in the areas of orbit determination and flight path/orbit control. The MRO navigation team has met the mission objectives and exceeded the project requirements, including the most stringent science requirements of all Mars orbiting missions. The experience offers a textbook example for future Mars missions.

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References

- ¹C.A. Halsell, et al., "Mars Reconnaissance Orbiter Aerobraking Control," AAS/AIAA Space Flight Mechanics Meeting, Sedona, AZ, January 28-February 1, 2007, AAS 07-243.
- ²J. Muckenthaler, "Mars Reconnaissance Orbiter Level 2.5 Project System Requirements, Rev. B," JPL D-22212, MRO 22-217, Jet Propulsion Laboratory, Pasadena, CA, Feb 2004.
- ³S.W. Demcak, et al., "Mars Reconnaissance Orbiter Orbit Determination During Aerobraking," AAS/AIAA Space Flight Mechanics Meeting, Sedona, AZ, January 28-February 1, 2007, AAS 07-245.
- ⁴S.M. Long, et. al., "Mars Reconnaissance Orbiter Aerobraking Daily Operations and Collision Avoidance," 20th International Symposium on Space Flight Dynamics Conference, Annapolis, MD, September 24-28, 2007.
- ⁵R.S. Bhat, et. al., "Mars Reconnaissance Orbiter Primary Science Orbit Acquisition," AAS/AIAA Astrodynamics Specialist Conference, Mackinac Island, MI, August 19-23, 2007.